Process Dynamics and Control

5. Frequency-Response Analysis & Design

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What are we up to?

Frequency-response analysis is the substitution $s = j\omega$: everything we need is the **magnitude** $|G(j\omega)|$ and the **phase** $\angle G(j\omega)$.

- Impose a **sinusoidal input** and measure the response — dynamics can be identified even **without a model**
- **Bode** and **Nyquist** plots are two pictures of the same functions of ω
- Gives a **relative** stability measure (gain and phase margins) and handles **dead time exactly**

Running examples

- Ex 5.10–5.13: inverse-response process $\dfrac{3(-2s+1)}{(10s+1)(5s+1)}$ under P and PI control
- Ex 4.4 revisited: $\dfrac{0.8e^{-2s}}{5s+1}$ — the dead-time case Routh could not do exactly

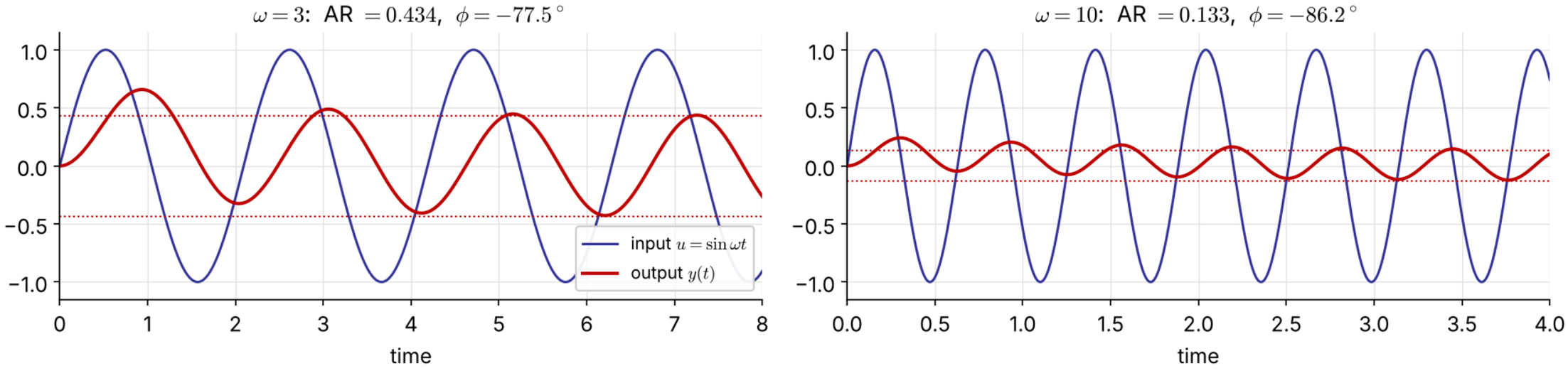
5.1 Magnitude and Phase Lag

A sine in, a sine out

First-order process $\tau_p \dfrac{dy}{dt} + y = K_p u, \quad G(s) = \dfrac{K_p}{\tau_p s + 1}, \quad u = A \sin \omega t$:

$$y(t) = \underbrace{\frac{AK_p\tau_p\omega}{\tau_p^2\omega^2 + 1} e^{-t/\tau_p}}_{\to\, 0} + \frac{AK_p}{\sqrt{\tau_p^2\omega^2 + 1}} \sin(\omega t + \phi), \qquad \phi = -\tan^{-1}(\omega\tau_p)$$

$G(s) = \dfrac{2}{1.5s + 1}$: after the transient, the output is a sine of the same frequency



Same frequency; the **amplitude ratio** $\mathrm{AR} = K_p / \sqrt{\tau_p^2\omega^2 + 1}$ falls and the **phase lag** grows as ω increases.

… and $G(j\omega)$ gives AR and ϕ directly

$$G(j\omega) = \frac{K_p}{j\omega\tau_p + 1} = \frac{K_p}{\tau_p^2\omega^2 + 1} - j\,\frac{K_p\omega\tau_p}{\tau_p^2\omega^2 + 1} = |G(j\omega)|\, e^{j\phi}$$

$$|G(j\omega)| = \frac{K_p}{\sqrt{\tau_p^2\omega^2 + 1}}, \qquad \phi = \angle G(j\omega) = -\tan^{-1}(\omega\tau_p)$$

General result (any **stable** G, see Appendix for the derivation): at large times

$$\frac{y_\infty(t)}{A} = |G(j\omega)|\, \sin\left(\omega t + \angle G(j\omega)\right)$$

No partial fractions needed — substitute $s = j\omega$ and take modulus and argument.

Products of transfer functions

For $G(s) = \dfrac{G_a(s)G_b(s)\cdots G_m(s)}{G_1(s)G_2(s)\cdots G_n(s)}$:

$$\log|G(j\omega)| = \sum_{\text{num}} \log|G_a(j\omega)| - \sum_{\text{den}} \log|G_i(j\omega)|, \qquad \angle G(j\omega) = \sum_{\text{num}} \angle G_a(j\omega) - \sum_{\text{den}} \angle G_i(j\omega)$$

On log scales, magnitudes **add** and phases **add** → a Bode plot is the **sum of simple pieces**.

Example 5.1 $G(s) = \dfrac{\tau_a s + 1}{(\tau_1 s + 1)(\tau_2 s + 1)}$

$$|G(j\omega)| = \frac{\sqrt{1+\omega^2\tau_a^2}}{\sqrt{1+\omega^2\tau_1^2}\,\sqrt{1+\omega^2\tau_2^2}}, \qquad \phi = \tan^{-1}(\omega\tau_a) - \tan^{-1}(\omega\tau_1) - \tan^{-1}(\omega\tau_2)$$

(The original slide left this blank — solution added.)

5.2 Graphical Analysis Tools

Bode and Nyquist plots

Bode plot — two plots against $\log \omega$

- magnitude: log–log ($|G|$ or $\mathbf{dB} = 20 \log_{10} |G|$)
- phase: semi-log, in **degrees**
- frequency in **rad per unit time**

Nyquist (polar) plot — $G(j\omega)$ drawn in the **complex G-plane** with ω as the parameter

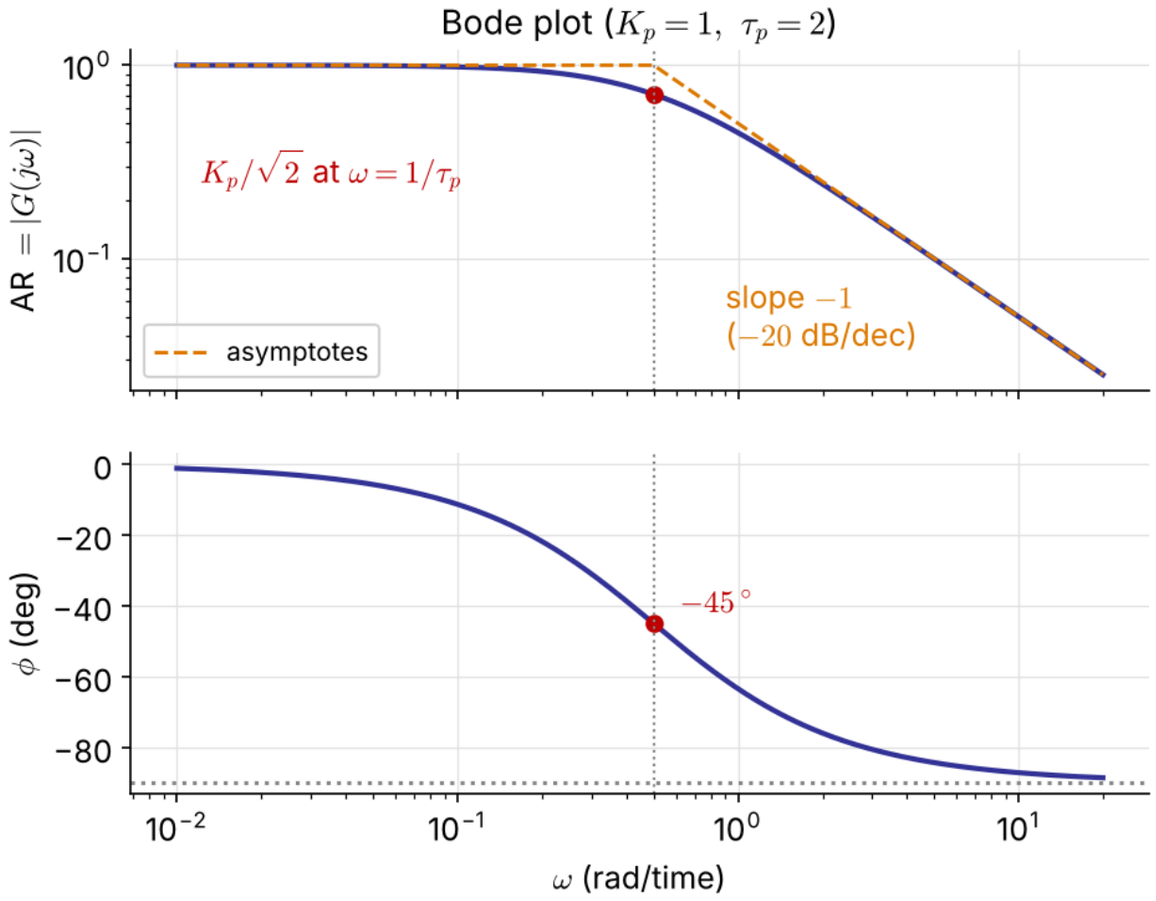
$$G(j\omega) = \operatorname{Re} G + j \operatorname{Im} G = |G| e^{j\phi}$$

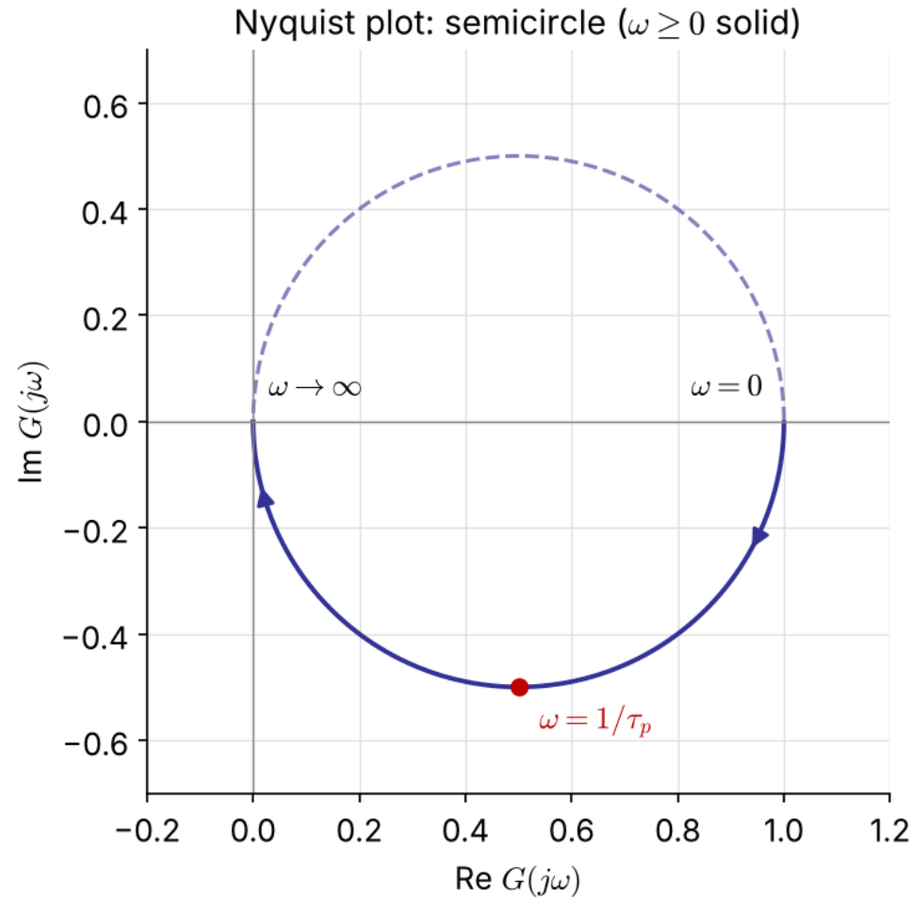
Same information; more compact, but ω is not shown explicitly.

AR	dB
0.01	−40
0.1	−20
0.5	−6
1	0
2	+6
10	+20
100	+40

A factor 10 is 20 dB; a factor 2 is ≈ 6 dB.

Example 5.2 First-order lag $K_p/(\tau_p s + 1)$





- $\omega \to 0$: $|G| = K_p,\ \phi = 0;\ \omega \to \infty$: $|G| \to K_p/\tau_p\omega,\ \phi \to -90^\circ$
- **Corner frequency** $\omega = 1/\tau_p$: $|G| = K_p/\sqrt{2},\ \phi = -45^\circ$
- **Bandwidth** ω_b: where $|G|$ drops to 0.707 of its low-frequency value ($= 1/\tau_p$ here)
- A **low-pass filter**: small τ ⇒ wide bandwidth ⇒ fast system

Example 5.3 Second-order $K/(\tau^2 s^2 + 2\zeta\tau s + 1)$

$$G(j\omega) = \frac{K\left[(1-\tau^2\omega^2) - j\,2\zeta\tau\omega\right]}{(1-\tau^2\omega^2)^2 + (2\zeta\tau\omega)^2}$$

$$|G| = \frac{K}{\sqrt{(1-\tau^2\omega^2)^2 + (2\zeta\tau\omega)^2}}$$

$$\phi = \tan^{-1}\frac{-2\zeta\tau\omega}{1-\tau^2\omega^2} \qquad (0 \to -180^\circ)$$

- $\omega \to 0$: $|G| = K$, $\phi = 0$
- $\omega = 1/\tau$: $|G| = K/2\zeta$, $\phi = -90^\circ$
- $\omega \to \infty$: $|G| \to K/\tau^2\omega^2$ (slope -2), $\phi \to -180^\circ$

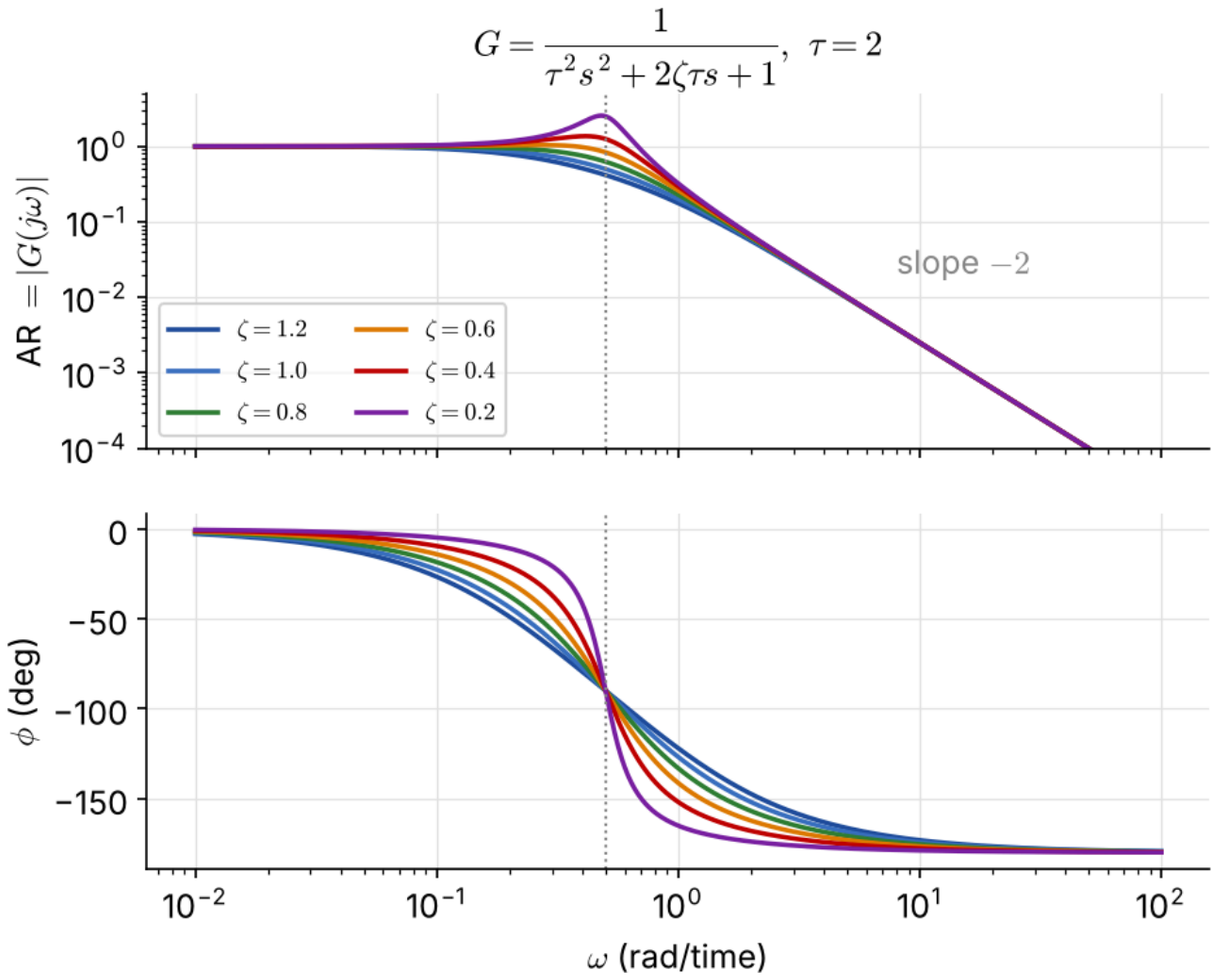
Resonance for $0 < \zeta < 1/\sqrt{2}$

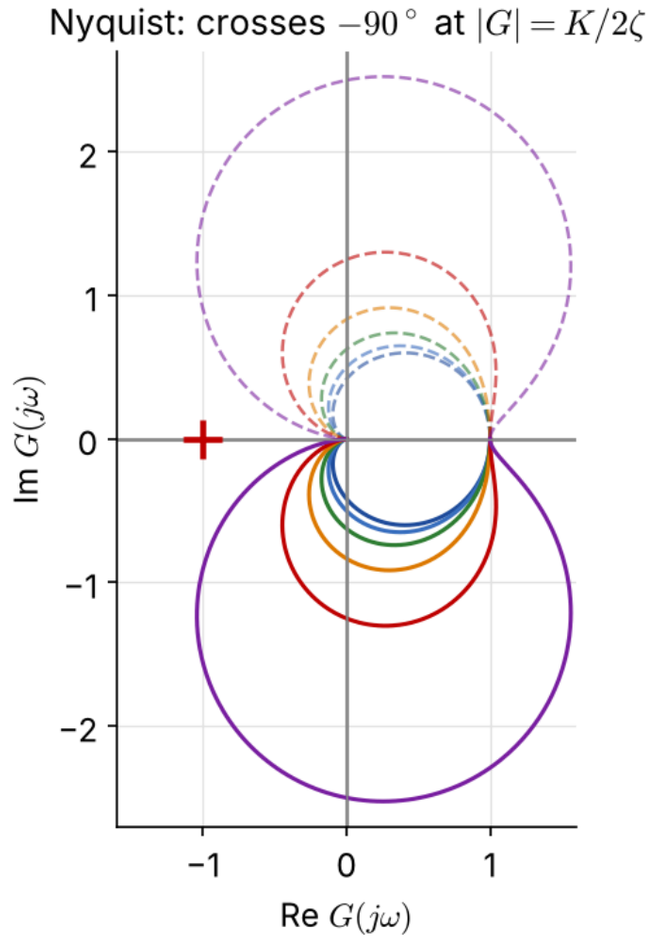
$$\omega_r = \frac{\sqrt{1-2\zeta^2}}{\tau}, \qquad M_{p\omega} = \frac{K}{2\zeta\sqrt{1-\zeta^2}}$$

Smaller ζ → higher peak → more overshoot in time (derivation: Appendix).

The original said the curve rises above K for $\zeta < 1/2$; that is the condition for $|G(j/\tau)| > K$. The resonant peak itself exists for $\zeta < 0.707$.

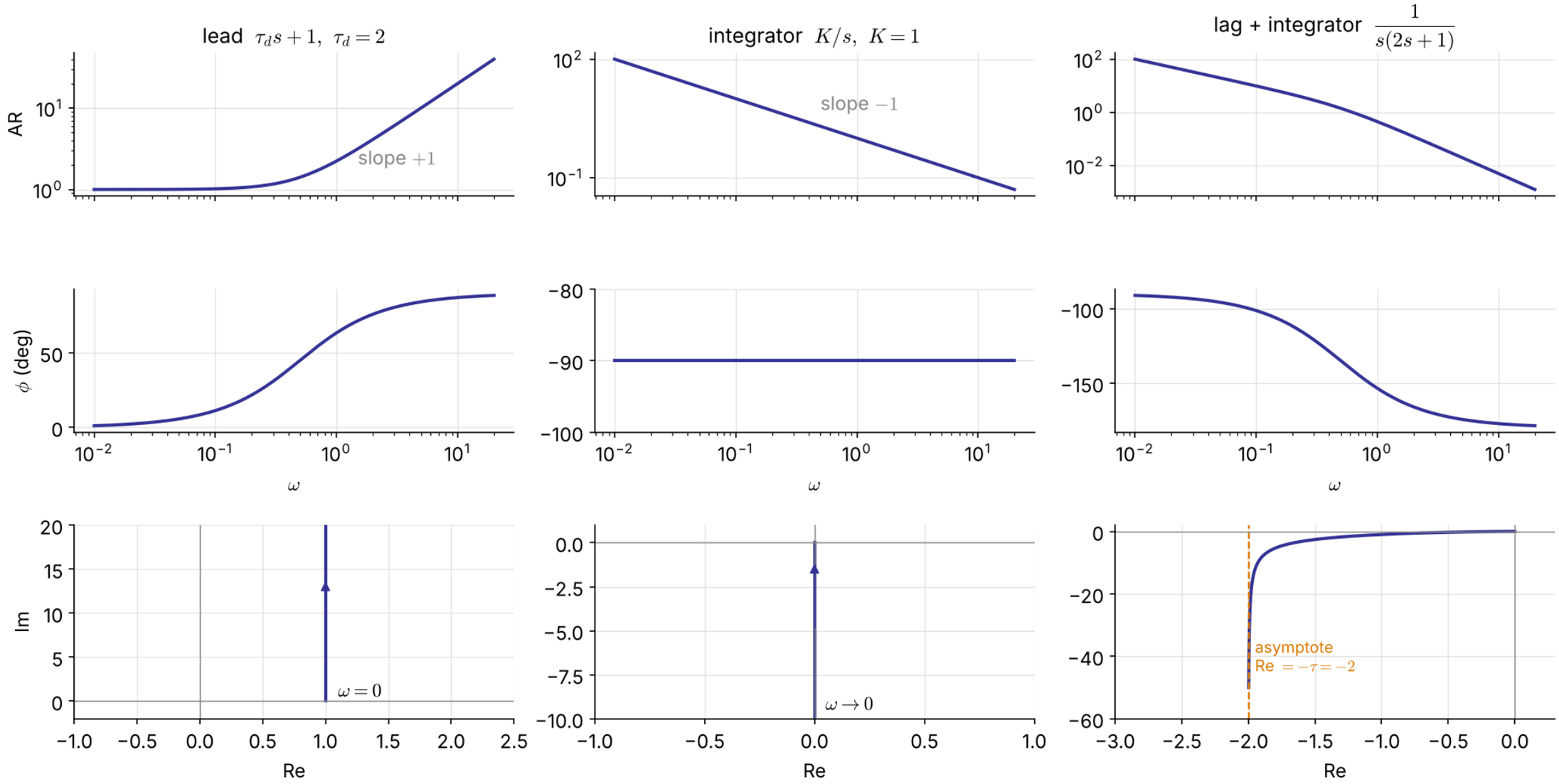
Example 5.3: Bode and Nyquist for $\zeta = 1.2 \ldots 0.2$





Low damping: a sharp **resonant peak** in $|G|$ and a fast phase drop at $\omega = 1/\tau$; the Nyquist curve bulges far beyond the radius-K circle.

Examples 5.4, 5.7, 5.8: lead, integrator, lag + integrator

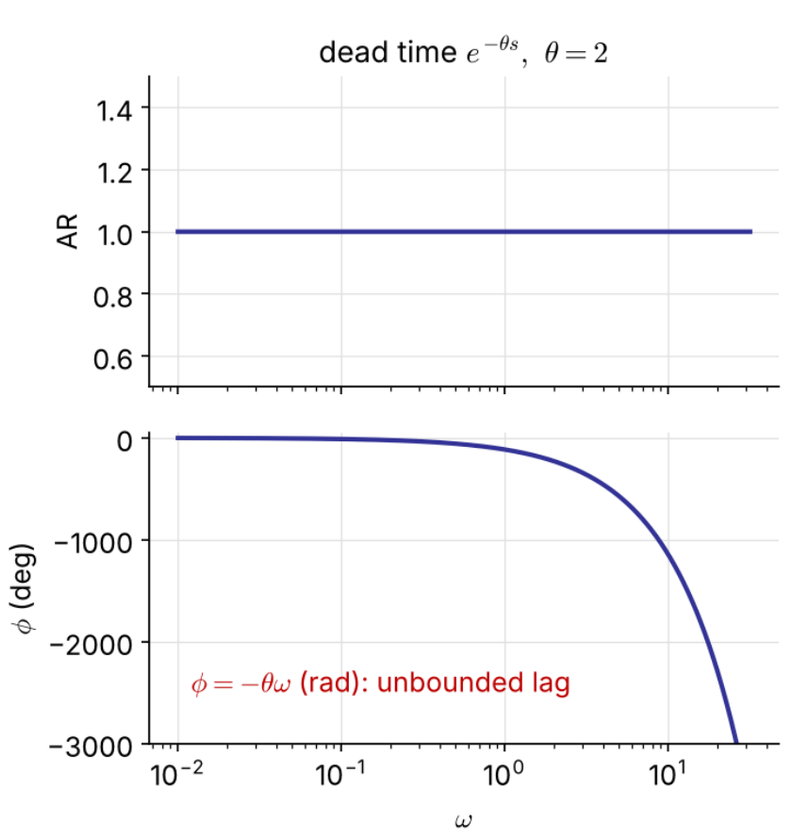


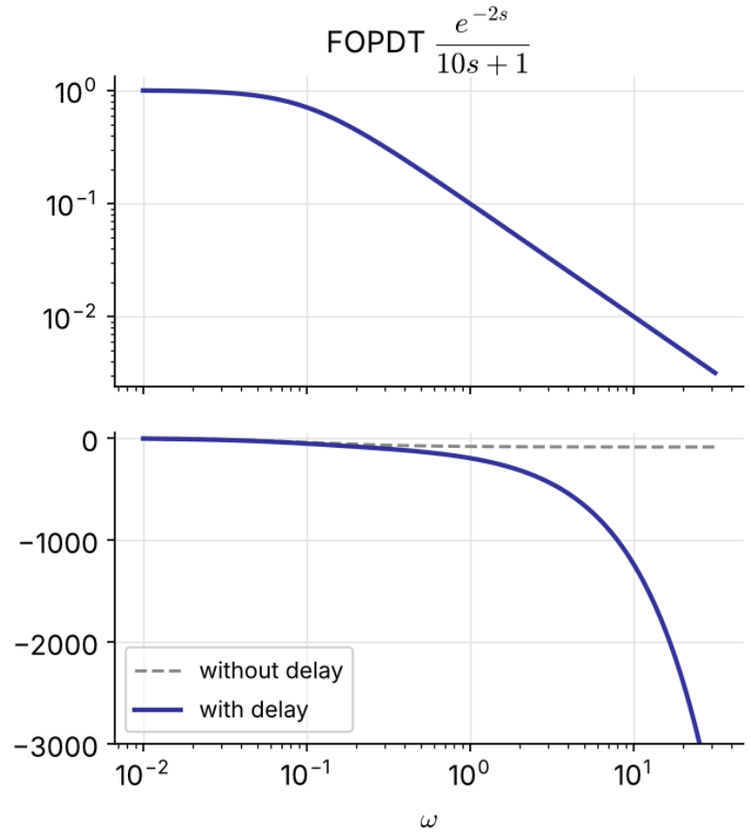
$|G| = \sqrt{1+\omega^2\tau_d^2},\ \phi = +\tan^{-1}\omega\tau_d$
(phase **lead**)

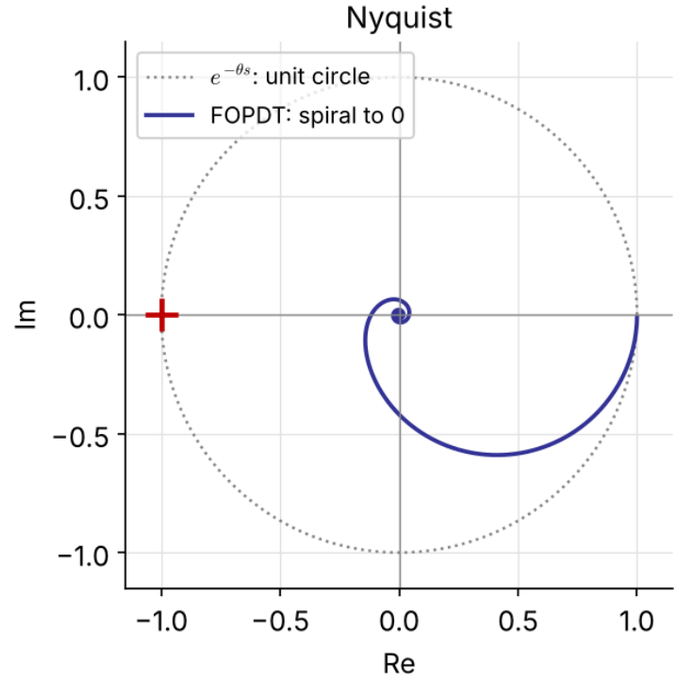
$|G| = K/\omega,\ \phi = -90°$ for all ω

ϕ from $-90°$ to $-180°$; Nyquist asymptote $\mathrm{Re} = -\tau$

Examples 5.5, 5.6: dead time







$|e^{-j\omega\theta}| = 1,\ \angle e^{-j\omega\theta} = -\omega\theta$ (rad): the phase lag **grows without bound**. FOPDT: $|G| = \dfrac{K_p}{\sqrt{1+\omega^2\tau_p^2}},\ \ \phi = -\tan^{-1}(\omega\tau_p) - \omega\theta.$

Nonminimum-phase systems contain dead time or RHP zeros (Ex 5.10); a **minimum-phase** system has neither, and no RHP poles.

Building blocks at a glance

Factor	AR $= \lvert G(j\omega)\rvert$	$\phi = \angle G(j\omega)$	high-ω slope	ϕ range
K	K	0	0	0
$\dfrac{1}{\tau s + 1}$	$\dfrac{1}{\sqrt{1+\omega^2\tau^2}}$	$-\tan^{-1}\omega\tau$	-1	$0 \to -90^\circ$
$\tau s + 1$ (LHP zero)	$\sqrt{1+\omega^2\tau^2}$	$+\tan^{-1}\omega\tau$	$+1$	$0 \to +90^\circ$
$-\tau s + 1$ (RHP zero)	$\sqrt{1+\omega^2\tau^2}$	$-\tan^{-1}\omega\tau$	$+1$	$0 \to -90^\circ$
$\dfrac{1}{s}$	$\dfrac{1}{\omega}$	-90°	-1	—
s	ω	$+90^\circ$	$+1$	—
$\dfrac{1}{\tau^2 s^2 + 2\zeta\tau s + 1}$	see Ex 5.3		-2	$0 \to -180^\circ$
$e^{-\theta s}$	1	$-\omega\theta$	0	unbounded

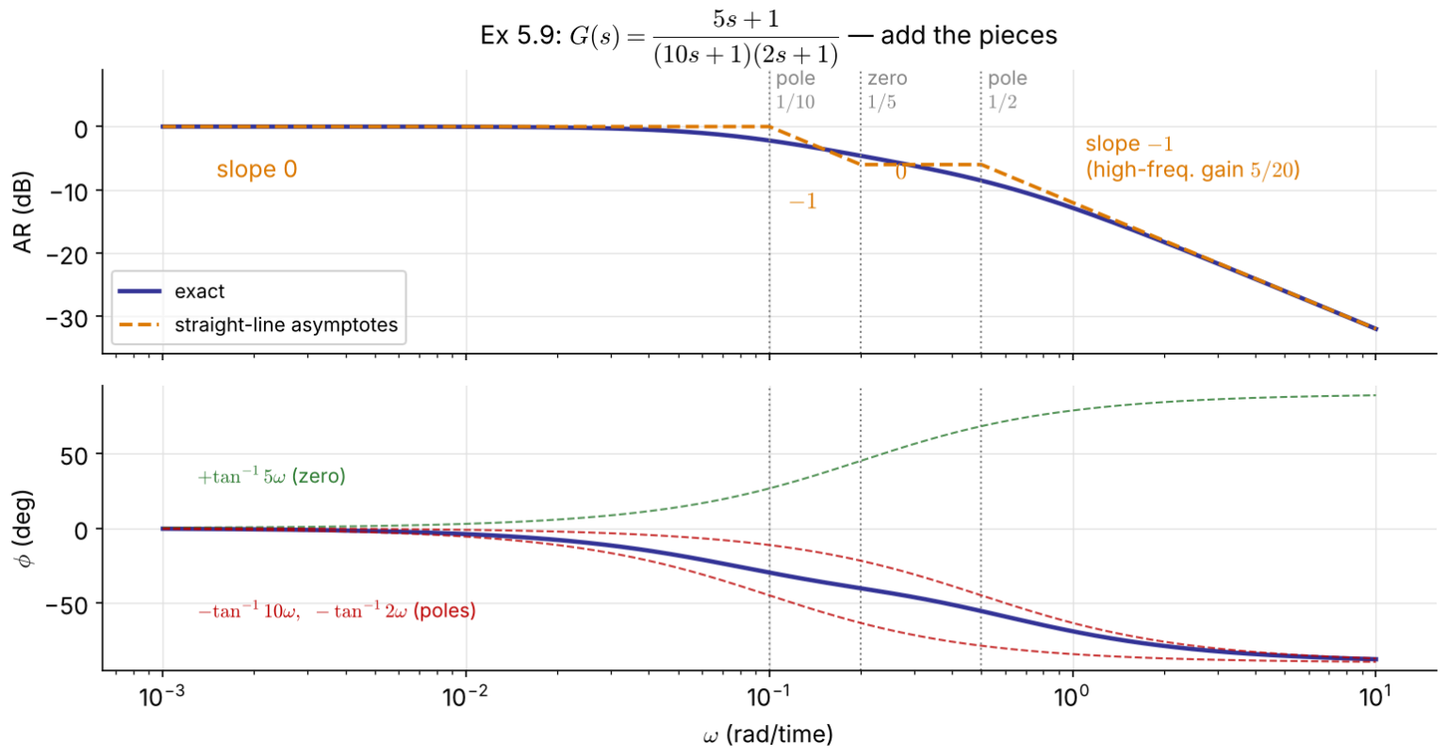
Slopes are in decades of AR per decade of ω (×20 for dB/dec). The RHP zero has the **magnitude** of an LHP zero but the **phase** of a pole — the origin of its trouble.

Sketching a Bode plot by asymptotes

1. Write G in **time-constant form** $K\dfrac{\prod(\tau_a s+1)}{s^q\prod(\tau_i s+1)}e^{-\theta s}$ and list the **corner frequencies** $1/\tau$.
2. **Low frequency**: $|G| \approx K/\omega^q$ — a line of slope $-q$ ($q=0$: horizontal at K).
3. Moving right, at each corner the slope changes by **−1 for a pole, +1 for a zero** (−2 for an underdamped pair).
4. **Phase**: each pole contributes $0 \to -90°$ (−45° at its corner), each zero $0 \to +90°$; dead time adds $-\omega\theta$.

For real poles and zeros the asymptotes are within 3 dB of the exact curve (largest error at a corner; underdamped pairs can deviate more near resonance) — good enough to read crossover frequencies and margins by hand.

Example 5.9 $(5s+1)/[(10s+1)(2s+1)]$ by asymptotes



Corners: 0.1 (pole), 0.2 (zero), 0.5 (pole)

ω range	slope
< 0.1	0
0.1–0.2	-1
0.2–0.5	0
> 0.5	-1

High frequency: $|G| \to \dfrac{5}{20\omega}$,
$\phi \to -90^\circ$

Phase:
$\tan^{-1} 5\omega - \tan^{-1} 10\omega - \tan^{-1} 2\omega$
never reaches -90° at finite ω

Bode and Nyquist in code

Python

```python
import numpy as np, control as ct, matplotlib.pyplot as plt

G = ct.tf(1, [2, 1])                              # Ex 5.2
ct.bode_plot(G, dB=False, deg=True)               # AR (log) and phase (deg)
ct.nyquist_plot(G)

# dead time: evaluate G(jw) e^{-jw*theta} exactly (no Pade)
w = np.logspace(-3, 2, 2000)
g = ct.tf(1, [10, 1])(1j*w) * np.exp(-2j*w)       # Ex 5.6, theta = 2
mag, phase = np.abs(g), np.degrees(np.unwrap(np.angle(g)))
fig, (a1, a2) = plt.subplots(2, 1, sharex=True)
a1.loglog(w, mag); a2.semilogx(w, phase)
```

MATLAB

```matlab
G = tf(1, [2 1]);                         % Ex 5.2
figure(1), bode(G)
figure(2), nyquist(G)

G6 = tf(1, [10 1], 'InputDelay', 2);      % Ex 5.6: exact dead time
figure(3), bode(G6)
figure(4), nyquist(G6)

K = 1; tau = 2;                           % Ex 5.3: family in zeta
for zeta = [1.2 1.0 0.8 0.6 0.4 0.2]
    bode(tf(K, [tau^2 2*zeta*tau 1])), hold on
end
```

5.3 Stability Analysis

Two frequency-domain tests

Bode stability criterion

G_{OL} strictly proper, **no poles on or right of the imaginary axis** (a single pole at the origin is allowed), and **one** phase crossover ω_c ($\angle G_{OL} = -180°$) and one gain crossover ω_g ($|G_{OL}| = 1$):

closed loop stable $\iff$ $|G_{OL}(j\omega_c)| < 1$

Applies with dead time — unlike Routh.

Nyquist stability criterion

G_{OL} proper, no unstable pole–zero cancellations.

- N = **clockwise** encirclements of $(-1, 0)$ by the Nyquist plot ($\omega : -\infty \to \infty$)
- P = open-loop poles in the RHP
- Z = closed-loop poles in the RHP

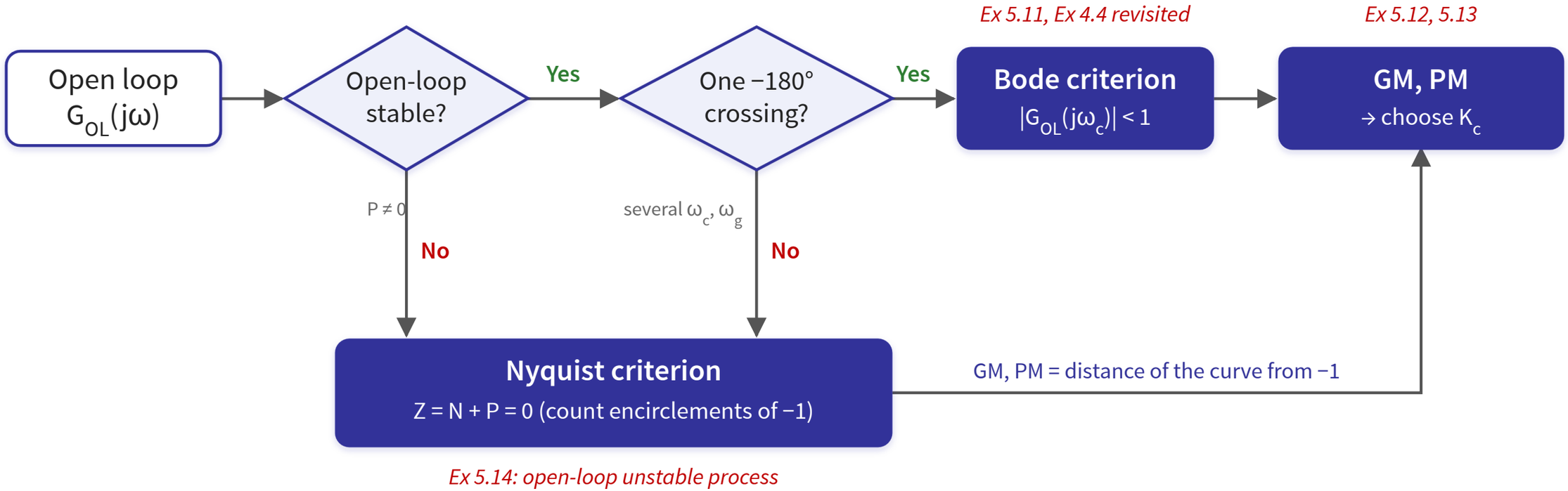
$$Z = N + P \qquad \textbf{stable} \iff Z = 0$$

(the original had $Z = N - P$; with clockwise N the sign is $+$)

Why -1? The characteristic equation $1 + G_{OL} = 0$ means $G_{OL} = -1$, i.e. $|G_{OL}| = 1$ **and** $\phi_{OL} = -180°$ — the **critical point**.

Reading the Nyquist criterion

- Most process loops are **open-loop stable**: $P = 0 \Rightarrow Z = N$ — unstable iff the curve **encircles** -1.
- A **counterclockwise** encirclement counts $N = -1$: each one can stabilize one open-loop unstable pole.
- Nyquist works where Bode does not: **open-loop unstable** processes, **several** crossover frequencies.



Ex 5.14: open-loop unstable process

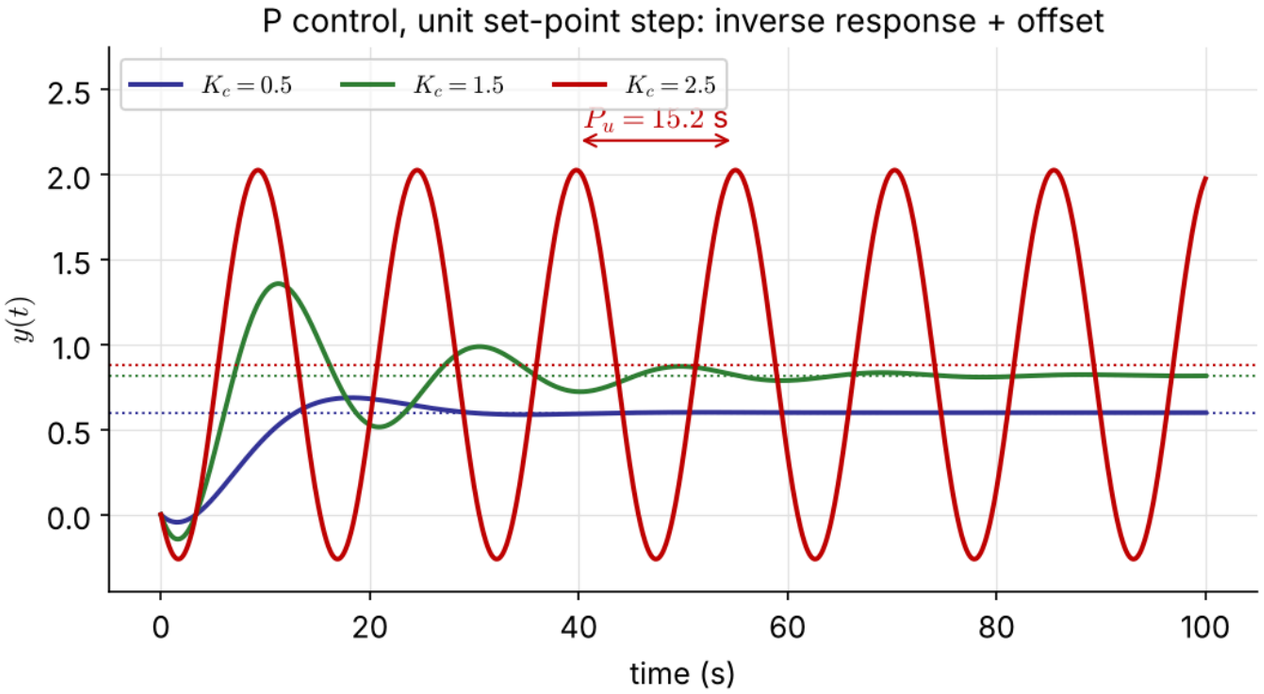
(a single pole at the origin is allowed in “open-loop stable”)

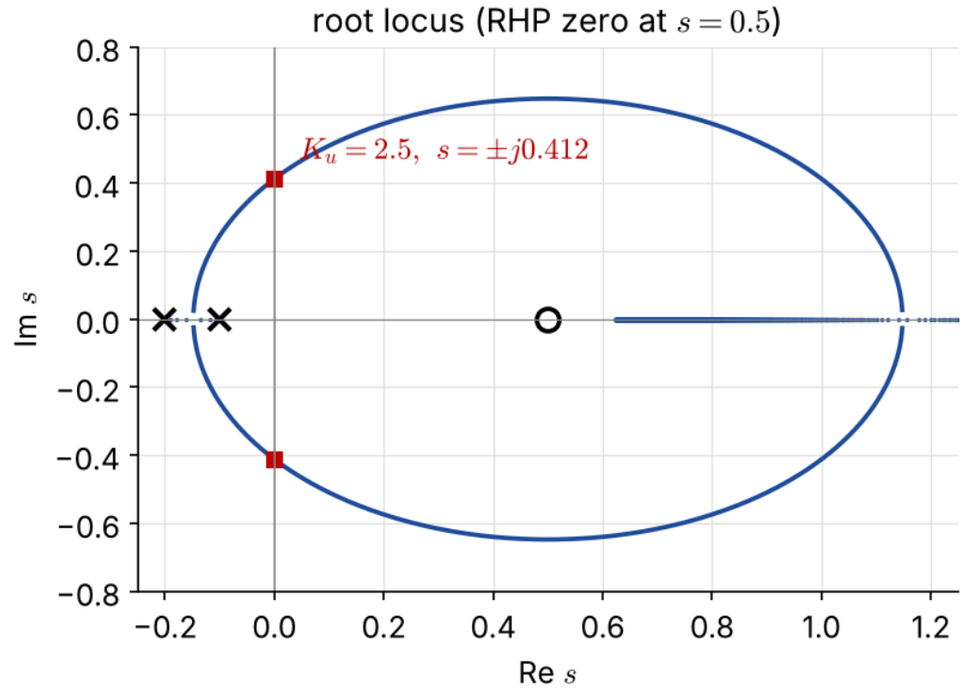
Examples 5.10, 5.11 Inverse-response process, P control

$G = \dfrac{3(-2s+1)}{(10s+1)(5s+1)}$ has a **RHP zero** at $s = 0.5$ → high gain is fundamentally limited.

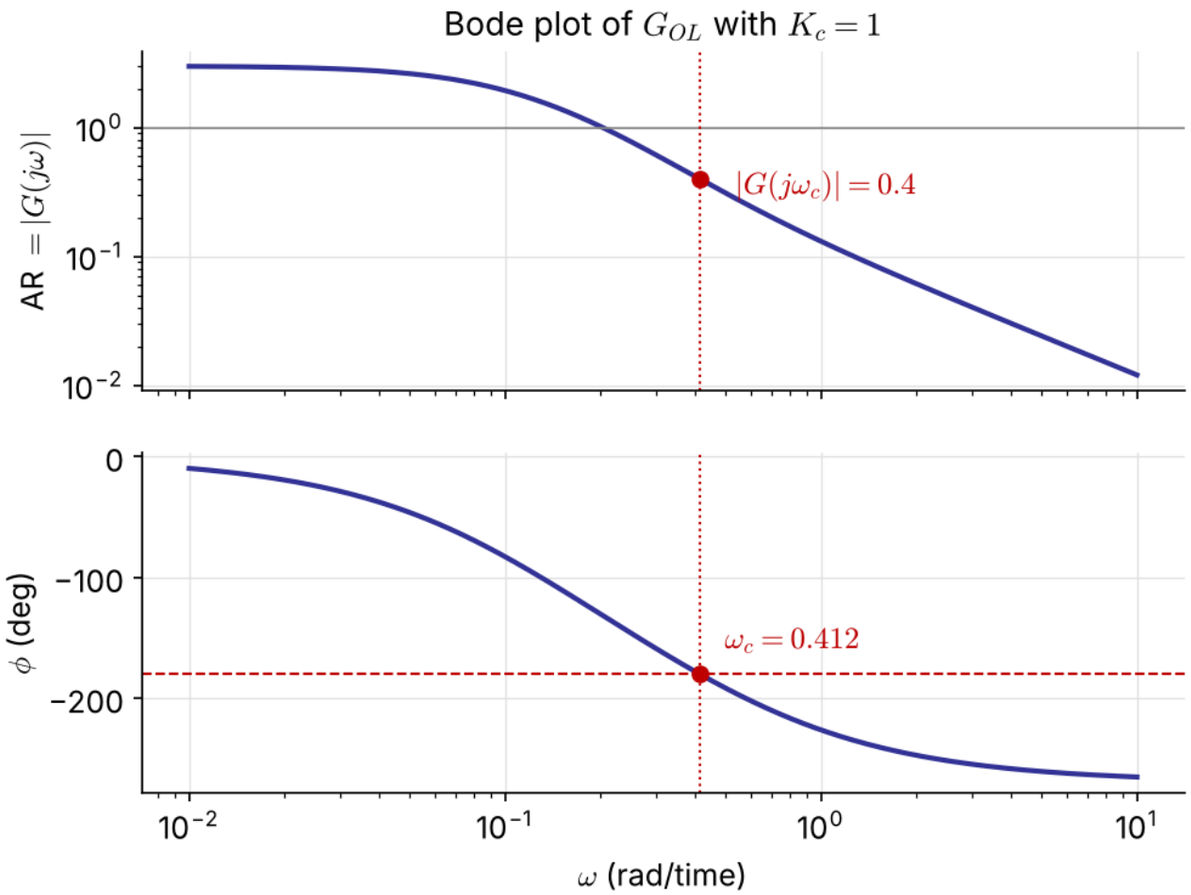
$$(10s+1)(5s+1) + 3K_c(-2s+1) = 0 \iff 50s^2 + (15 - 6K_c)s + (1 + 3K_c) = 0$$

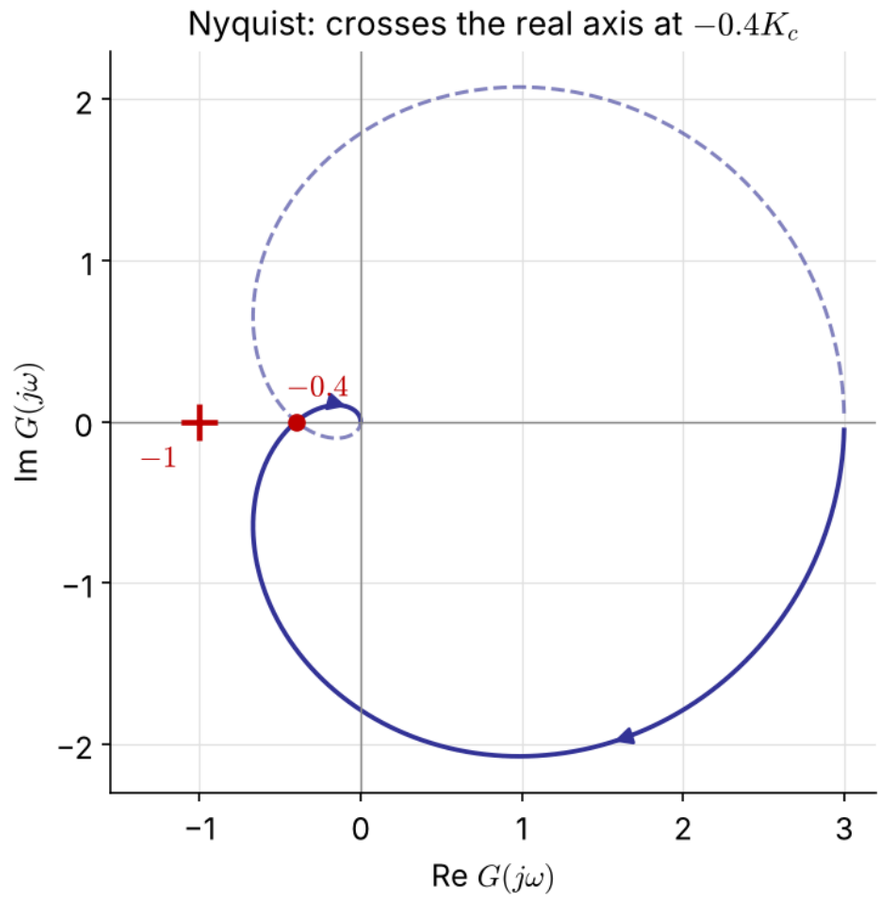
Routh: $15 - 6K_c > 0,\ 1 + 3K_c > 0 \Rightarrow -\frac{1}{3} < K_c < 2.5$ → **ultimate gain $K_u = 2.5$**; at K_u: $50s^2 + 8.5 = 0, s = \pm j0.412$





Example 5.11 by frequency response



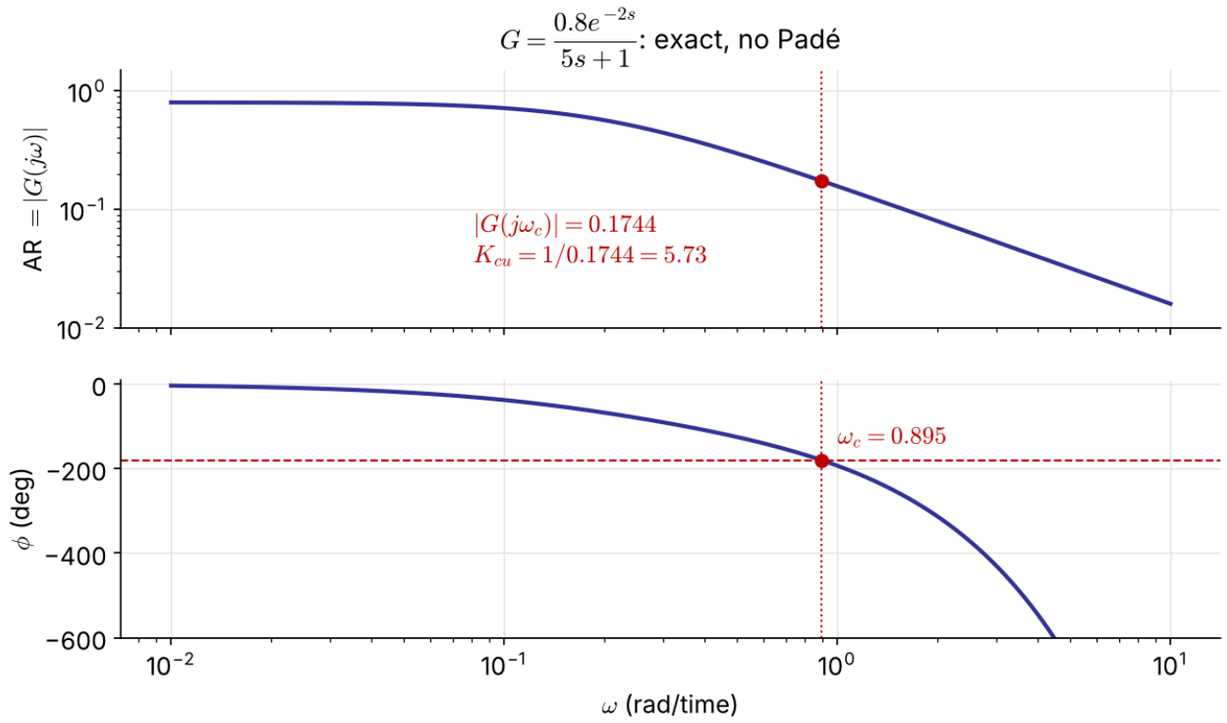


With $K_c = 1$: phase crossover $\omega_c = 0.412$, $|G(j\omega_c)| = 0.4$

$|G_{OL}(j\omega_c)| = 0.4K_c < 1 \iff K_c < 2.5$ — the same as Routh

Nyquist: the real-axis crossing at $-0.4K_c$ reaches -1 when $K_c = 2.5$; $P_u = 2\pi/\omega_c = 15.2$ s

Ex 4.4 revisited: dead time handled exactly



$$G_{OL} = K_c \frac{0.8e^{-2s}}{5s+1}$$

Phase crossover: $-\tan^{-1}(5\omega_c) - 2\omega_c = -\pi$

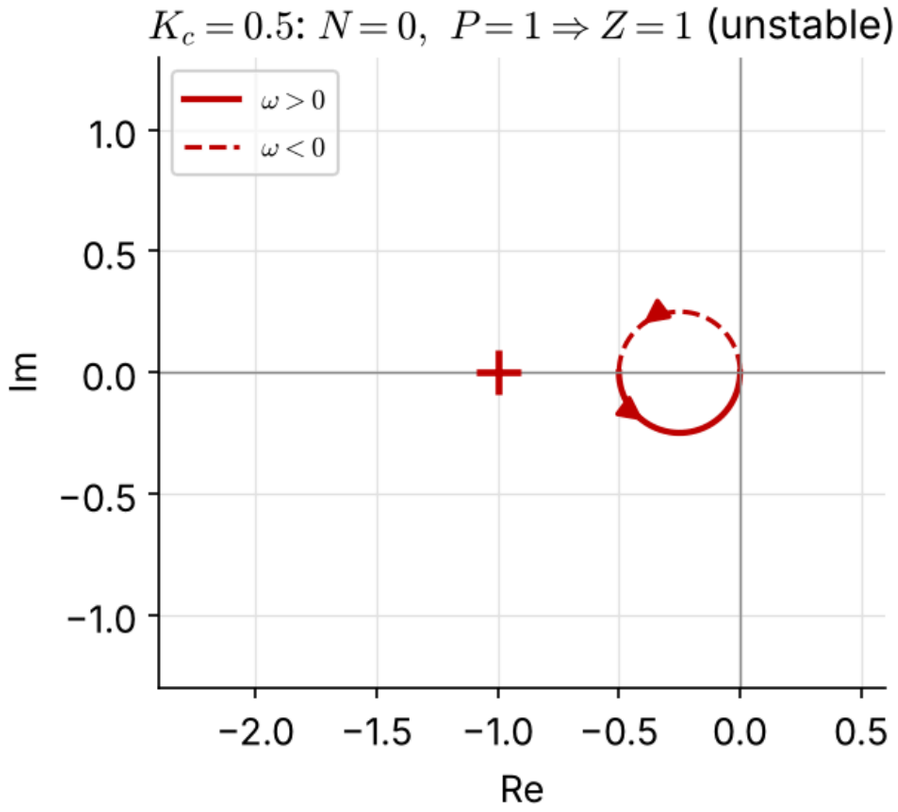
$\Rightarrow \omega_c = 0.895$

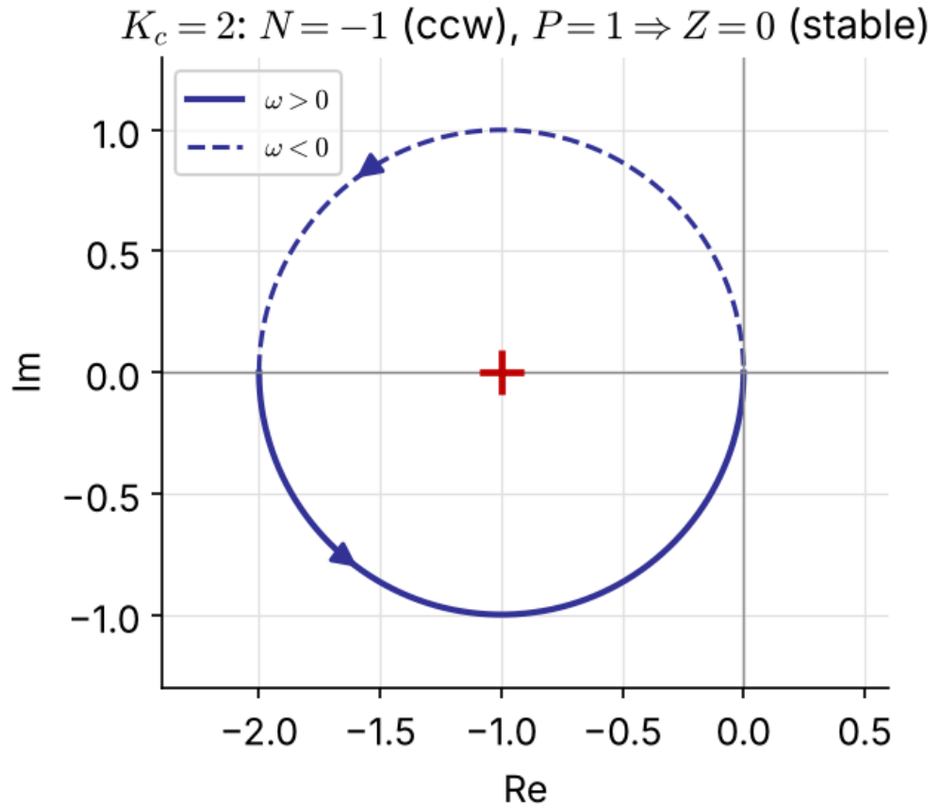
$$K_{cu} = \frac{1}{|G(j\omega_c)|} = \frac{\sqrt{1+25\omega_c^2}}{0.8} = 5.73$$

The **same** answer as direct substitution in Ch 4 — read off one Bode plot, with no Padé approximation (1st-order Padé gave 7.5).

Example 5.14 Nyquist for an open-loop unstable process

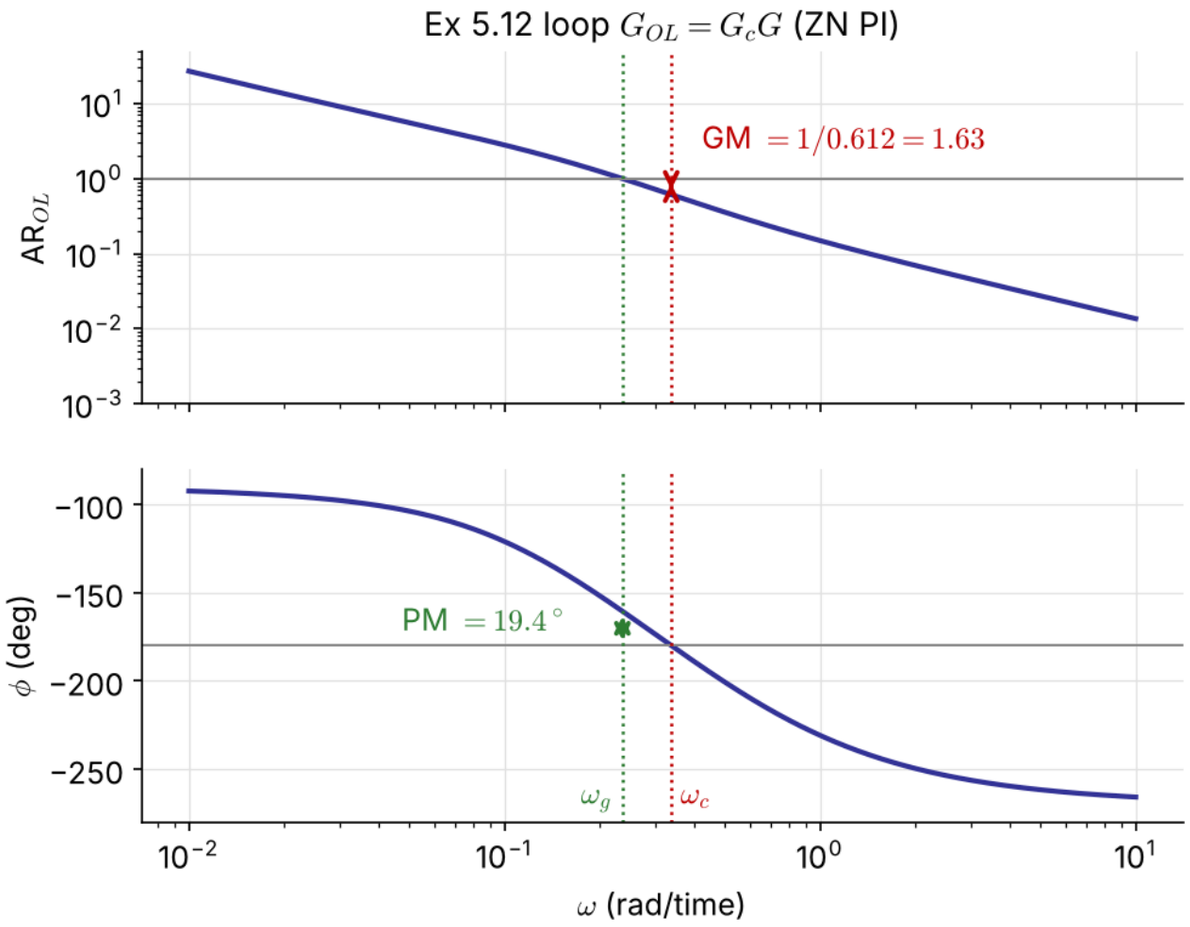
$$G_{OL} = \frac{K_c}{s-1} \text{ (Ex 4.1 with } K = a = 1)$$

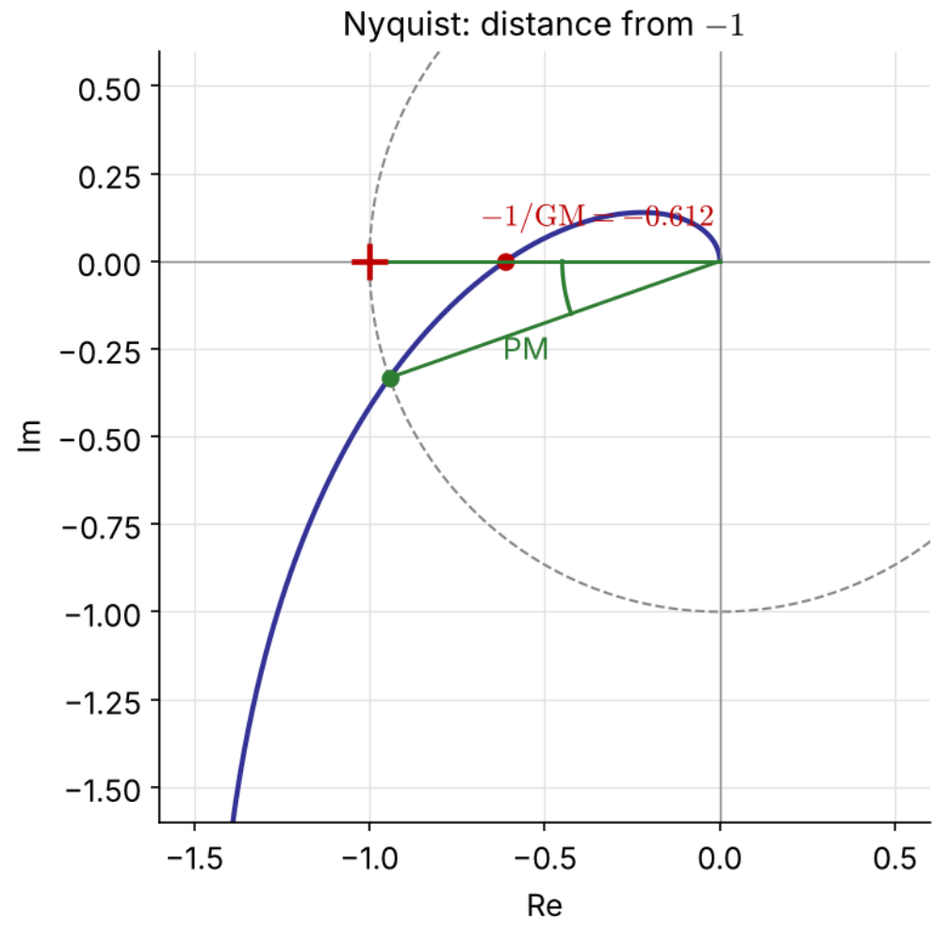




$K_c/(s-1)$ is a circle of diameter K_c through 0 and $-K_c$, traversed **counterclockwise**. $P = 1$, so stability needs $N = -1$: the circle must enclose $-1 \Rightarrow K_c > 1$ — exactly Ex 4.1 ($K_c K > a$). The Bode criterion cannot be used here ($P \neq 0$).

Gain and phase margins





Gain margin (at ω_c, where $\angle G_{OL} = -180°$)

$$\mathrm{GM} = \frac{1}{|G_{OL}(j\omega_c)|}$$

Phase margin (at ω_g, where $|G_{OL}| = 1$)

$$\mathrm{PM} = 180° + \angle G_{OL}(j\omega_g)$$

What the margins protect against

GM — factor by which the loop gain can grow before instability → safeguard against **gain (model) error**

Typical spec: GM > 2 (6 dB)

GM $= 1/|G_{OL}(j\omega_c)|$ is the distance from -1 along the real axis

PM — extra phase lag tolerated at ω_g → safeguard against **dead-time error**

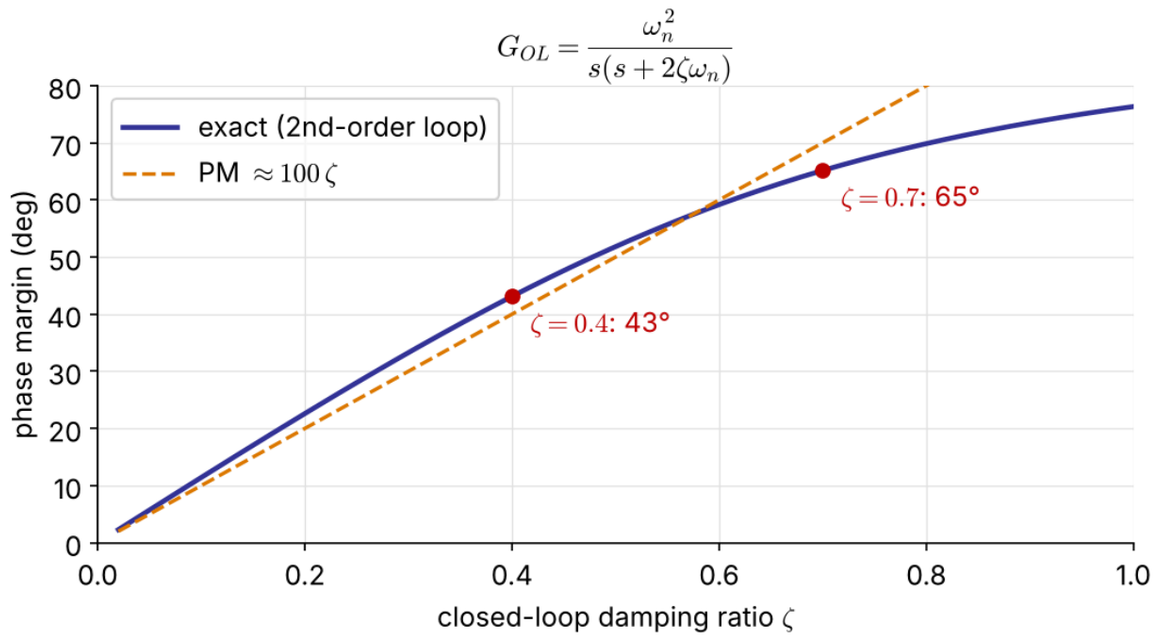
Typical spec: PM > 30° (often 45°)

Extra delay that destabilizes:

$$\theta_{\max} = \frac{\text{PM (rad)}}{\omega_g}$$

- If several crossings: take the one closest to -1 (largest $|G_{OL}|$) for GM, the smallest PM. Low-order minimum-phase plants may have GM $= \infty$.
- Lowering ω_g (smaller bandwidth, slower loop) tolerates larger delay errors. Ex 5.12: $\theta_{\max} = 0.339/0.236 = 1.4$ s.

Phase margin and damping



For a loop that behaves like $\dfrac{\omega_n^2}{s(s+2\zeta\omega_n)}$:

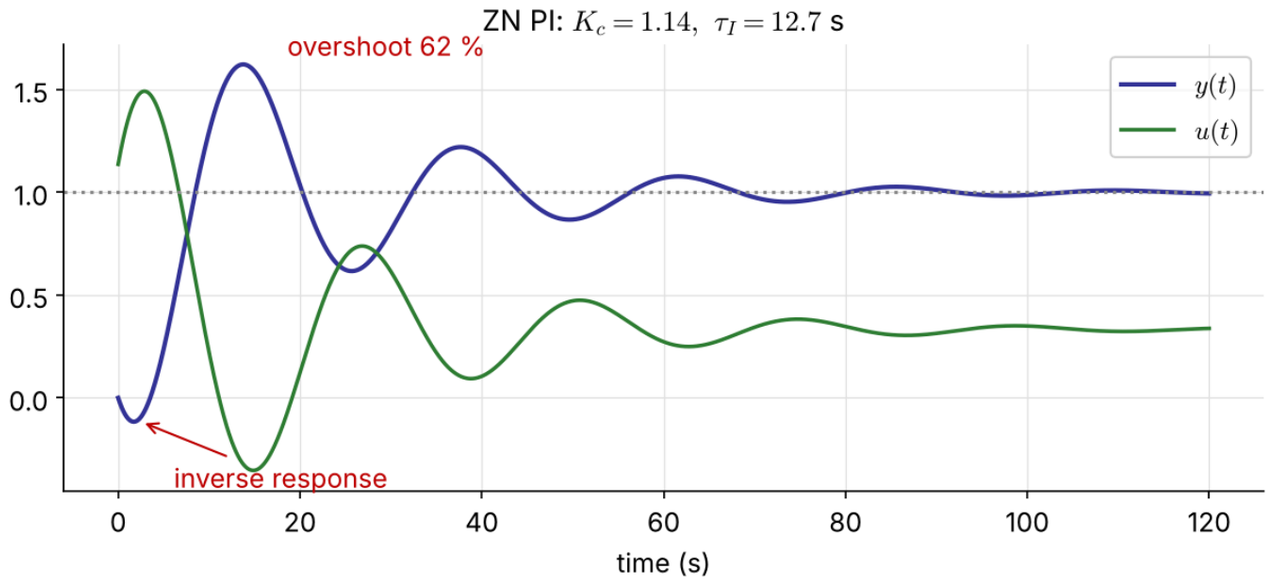
$$\mathrm{PM} = \tan^{-1}\frac{2\zeta}{\sqrt{\sqrt{1+4\zeta^4}-2\zeta^2}} \;\approx\; 100\,\zeta\ (\mathrm{deg})$$

PM 45° $\leftrightarrow$ $\zeta \approx 0.4$ (overshoot ≈ 25 %)

PM 65° $\leftrightarrow$ $\zeta \approx 0.7$ (overshoot ≈ 5 %)

A small PM means an oscillatory closed loop — the bridge to the ζ-based root-locus design of Ch 4.

Example 5.12 Ziegler–Nichols PI for the inverse-response process



$G_c = K_c\left(1 + \dfrac{1}{\tau_I s}\right)$ removes the P-control offset.

ZN settings from Ex 5.11 ($K_u = 2.5,\ P_u = 15.2$ s):

$$K_c = \frac{K_u}{2.2} = 1.14, \qquad \tau_I = \frac{P_u}{1.2} = 12.7\ \text{s}$$

GM $= 1.63$, PM $= 19.4°$
— **too small** (GM > 2, PM > 30°)

→ 62 % overshoot, slow-decaying oscillation

ZN tunings are deliberately “aggressive” (quarter-decay); use them only as a starting point.

Margins in code

Python

```python
import numpy as np, control as ct

G = ct.tf([-6, 3], np.polymul([10, 1], [5, 1]))
gm, pm, wc, wg = ct.margin(G)          # P control with Kc = 1
Ku, Pu = gm, 2*np.pi/wc                # 2.5, 15.2 s

Kc, tI = Ku/2.2, Pu/1.2                # Ziegler-Nichols PI
L = Kc*ct.tf([tI, 1], [tI, 0])*G
gm, pm, wc, wg = ct.margin(L)          # GM 1.63, PM 19.4 deg
theta_max = np.radians(pm)/wg          # 1.43 s of extra dead time
ct.bode_plot(L, display_margins=True)
```

MATLAB

```matlab
G = tf([-6 3], conv([10 1],[5 1]));
[Ku,~,wc] = margin(G);   Pu = 2*pi/wc;      % 2.5, 15.2 s

Kc = Ku/2.2;   tI = Pu/1.2;                 % Ziegler-Nichols PI
L = Kc*tf([tI 1],[tI 0])*G;
[GM,PM,wc,wg] = margin(L)                   % 1.63, 19.4 deg
theta_max = PM*pi/180/wg                    % 1.43 s
margin(L)                                   % Bode plot with margins marked
```

5.4 Controller Design

Choosing K_c without trial and error

Write the loop as $1 + G_{OL} = 1 + K_c G^* = 0$, where G^* contains **everything except K_c** (process, valve, sensor, and the τ_I, τ_D terms of the controller).

From a GM spec

$\angle K_c = 0$, so G^* and G_{OL} share ω_c:

$$K_{cu} = \frac{1}{|G^*(j\omega_c)|}, \qquad K_c = \frac{K_{cu}}{\mathrm{GM}}$$

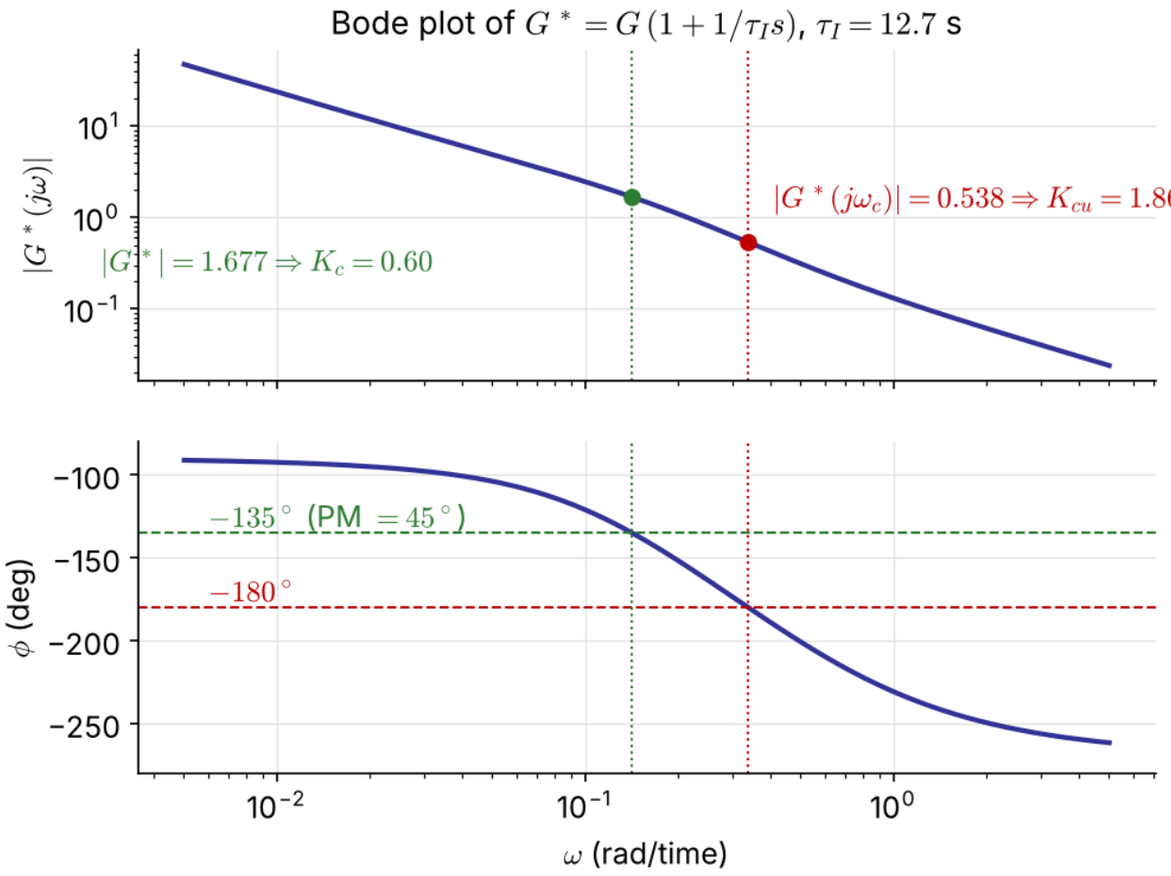
From a PM spec

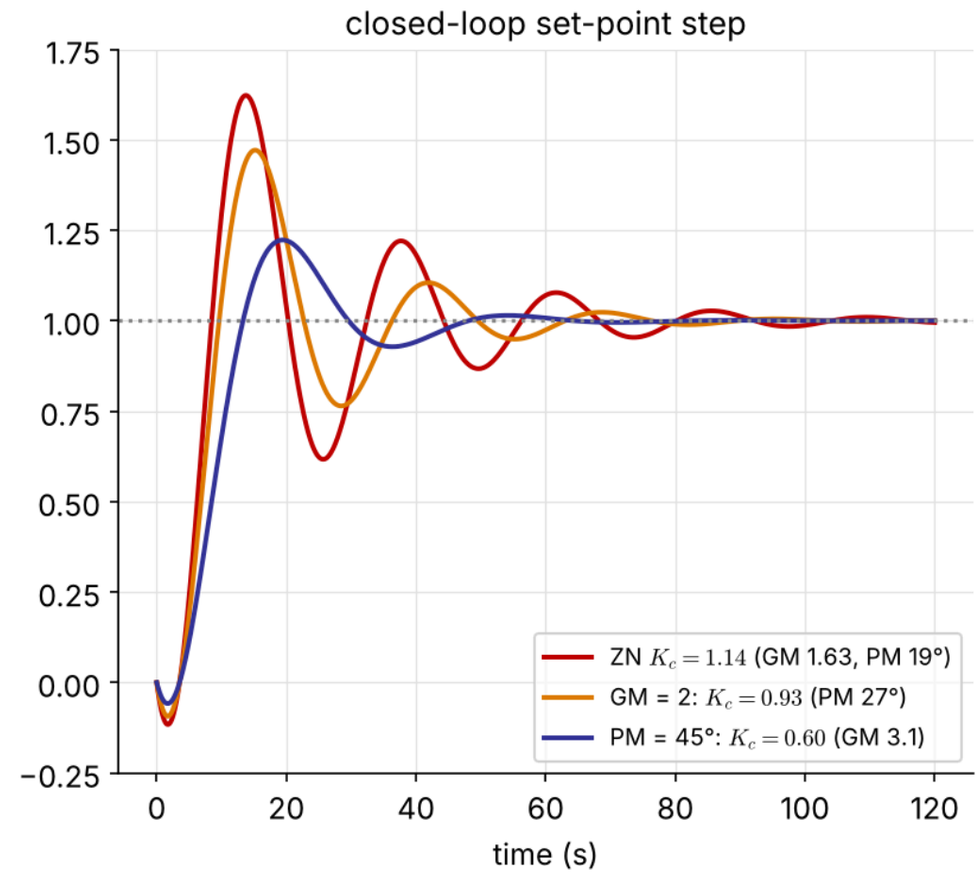
Find ω_g where $\angle G^* = -180^\circ + \mathrm{PM}$ (e.g. -135° for 45°):

$$K_c = \frac{1}{|G^*(j\omega_g)|}$$

Compute both and take the **smaller K_c**, which satisfies both specs. One Bode plot of G^* does it all.

Example 5.13 Retuning Ex 5.12 for GM ≥ 2 and PM ≥ 45°





Keep $\tau_I = 12.7$ s: $G^* = G(1 + 1/\tau_I s)$, $\omega_c = 0.338$, $K_{cu} = 1.86$

- GM $= 2 \rightarrow K_c = 0.93$ (but PM only 27°)
- PM $= 45^\circ \rightarrow \omega_g = 0.142$, $K_c = 0.60$ (GM 3.1)

$K_c = 0.60$ meets **both** specs: overshoot 62 % → 22 %, settles in 45 s instead of 89 s

Example 5.13 in code

Python

```python
tI = Pu/1.2
Gs = ct.tf([tI, 1], [tI, 0])*G                  # G* = everything except Kc

Kcu = ct.margin(Gs)[0]                          # 1.86
Kc_GM = Kcu/2                                   # GM = 2   -> 0.93

w = np.logspace(-3, 1, 100000)
g = Gs(1j*w)
phase = np.degrees(np.unwrap(np.angle(g)))
i = np.argmin(abs(phase - (-180 + 45)))         # phase = -135 deg
Kc_PM = 1/abs(g[i])                             # PM = 45 deg -> 0.60

Kc = min(Kc_GM, Kc_PM)
print(ct.margin(Kc*Gs), ct.step_info(ct.feedback(Kc*Gs, 1)))
```

MATLAB

```matlab
tI = Pu/1.2;
Gs = tf([tI 1],[tI 0])*G;                       % G* = everything except Kc

Kcu = margin(Gs);   Kc_GM = Kcu/2;              % 1.86 -> 0.93

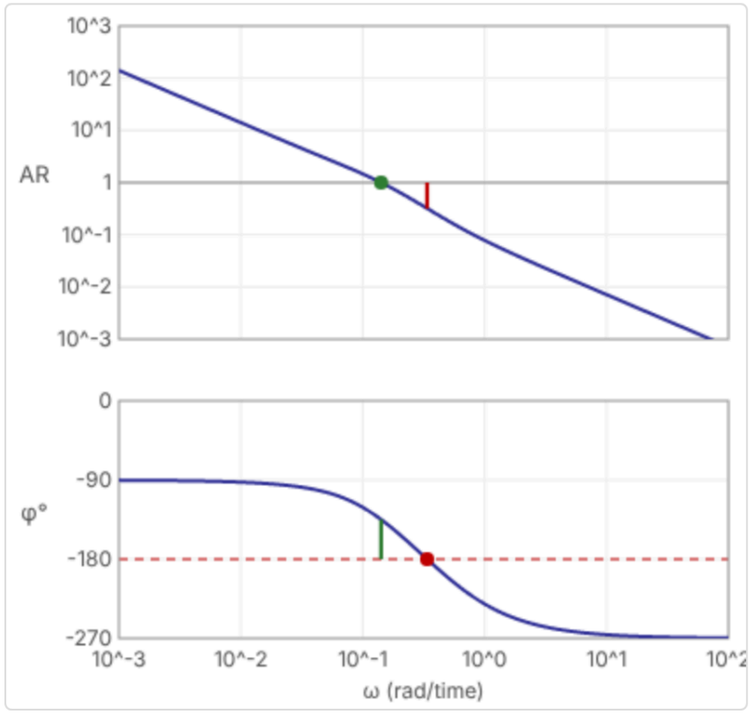
w = logspace(-3, 1, 1e5);
[m, p] = bode(Gs, w);   m = squeeze(m);   p = squeeze(p);
[~, i] = min(abs(p + 135));                     % phase = -135 deg
Kc_PM = 1/m(i)                                  % 0.60

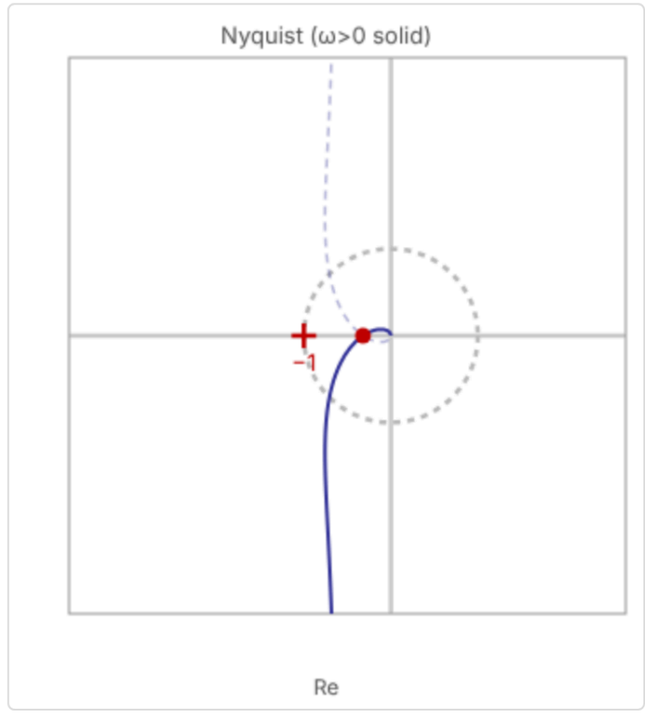
Kc = min(Kc_GM, Kc_PM);
margin(Kc*Gs), stepinfo(feedback(Kc*Gs, 1))
```

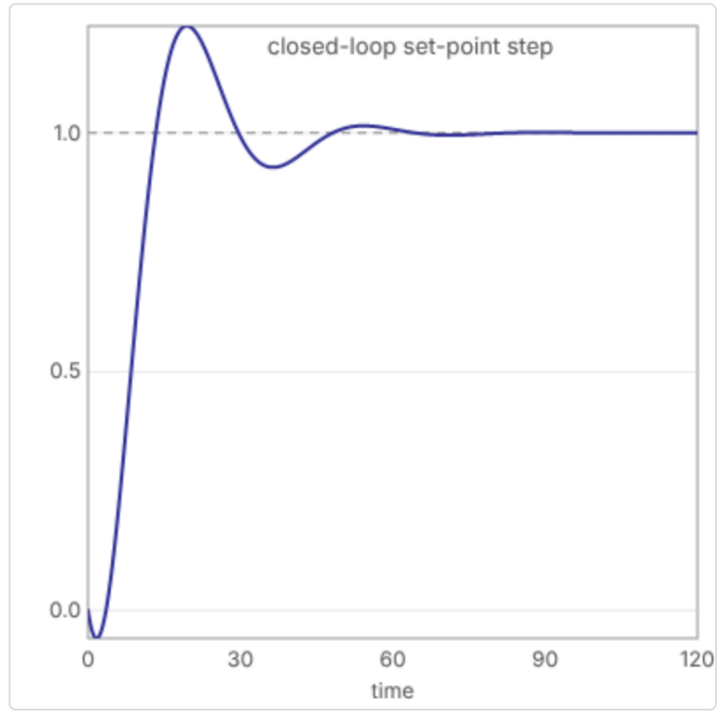
Interactive: move K_c, watch the margins

System Ex 5.12 — PI (τI=12.7), same plant K c = **0.595** STABLE

GM = 3.122 (ω_c = 0.338) **PM = 45.1°** (ω_g = 0.142) $K_{cu} = K_c \cdot GM$ = 1.858







Interactive version: seoultechpse.github.io/ProCon/ch5/ch5_frequency.html — move K_c for Ex 5.11, 5.12, Ex 4.4 and Ex 4.2.

A final word: frequency response or root locus?

They **complement** each other; a design decision is often made only after both.

Root locus (Ch 4)

- shows the **transient response** and how it moves with K_c
- needs a fairly accurate model
- does not handle dead time well

Frequency response (Ch 5)

- **relative** stability: GM, PM; dead time **exactly**
- Bode plot can come from **experiments** (sine or pulse tests) — no model needed, or a check of one
- several crossovers: use Nyquist, not Bode alone

Task	MATLAB	Python (`control`)
Bode / Nyquist	`bode(G)`, `nyquist(G)`	`ct.bode_plot(G)`, `ct.nyquist_plot(G)`
dead time	`tf(..., 'InputDelay', θ)`	`G(1j*w)*np.exp(-1j*w*θ)`
margins	`[GM,PM,wc,wg] = margin(L)`	`gm, pm, wc, wg = ct.margin(L)`

Problem #2 (homework)

$$G_p = \frac{2e^{-s}}{10s+1}, \qquad G_c = K_c\left(1+\frac{1}{\tau_I s}\right), \quad \tau_I = 10$$

a. Show that $\tau_I = 10$ makes G_{OL} a pure integrator with dead time. Sketch its Bode plot by hand.

b. Find the phase crossover ω_c and the ultimate gain K_{cu} analytically.

c. Find K_c for PM $= 45^\circ$. What is the resulting GM? Comment on the result.

d. Check (b)–(c) with `ct.margin` and simulate the set-point response.

Hint: the phase of $e^{-j\omega}/j\omega$ is $-\pi/2 - \omega$.

Resources

- **Slides (interactive HTML):** seoultechpse.github.io/ProCon/ch5/ch5_frequency.html
- **PDF:** slides · handout with note space
- **Python notebook:** Open in Colab — every calculation and figure of this chapter

Appendix

General frequency response: derivation

Stable $G = Q/P$, input $A\sin\omega t$: $\displaystyle Y(s) = \frac{Q(s)}{P(s)}\,\frac{A\omega}{(s+j\omega)(s-j\omega)}$

All roots of P have negative real parts, so their terms decay. At large times only the input poles remain:

$$Y_\infty(s) = \frac{a}{s+j\omega} + \frac{a^*}{s-j\omega}, \qquad a = (s+j\omega)Y(s)\big|_{s=-j\omega} = \frac{AG(-j\omega)}{-2j}$$

With $G(\pm j\omega) = |G(j\omega)|e^{\pm j\phi}$:

$$\frac{y_\infty(t)}{A} = |G(j\omega)|\left(\frac{e^{-j(\omega t+\phi)}}{-2j} + \frac{e^{j(\omega t+\phi)}}{2j}\right) = |G(j\omega)|\sin(\omega t+\phi), \qquad \phi = \angle G(j\omega)$$

Resonance of the second-order system

$$\frac{d}{d\omega}\left[(1-\tau^2\omega^2)^2+(2\zeta\tau\omega)^2\right]=0 \;\Rightarrow\; \tau^2\omega_r^2=1-2\zeta^2$$

$$M_{p\omega}=|G(j\omega_r)|=\frac{K}{\sqrt{(2\zeta^2)^2+4\zeta^2(1-2\zeta^2)}}=\frac{K}{2\zeta\sqrt{1-\zeta^2}}$$

A real ω_r requires $1-2\zeta^2>0$, i.e. $\zeta<1/\sqrt{2}$.

ζ	0.2	0.4	0.6
$\tau\omega_r$	0.96	0.82	0.53
$M_{p\omega}/K$	2.55 (8.1 dB)	1.36 (2.7 dB)	1.04 (0.4 dB)